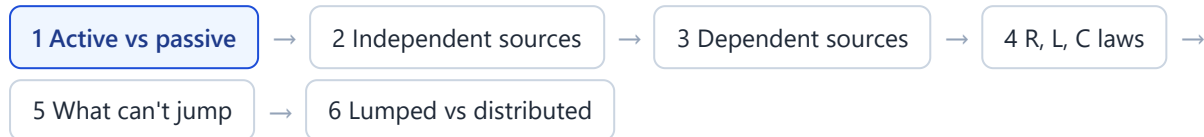


01 — Circuit elements and sources

Class notes p.1–3, set 2 p.1–3 · MTE syllabus item 1 · needed by every other file · no board question yet: the openers here are Predict prompts, not exam questions

Map



The professor's classification (class p.1):

Element	Kind	Examples
Active · independent	fixed value	voltage source, current source
Active · dependent	value set by another circuit quantity	VCVS, VCCS, CCVS, CCCS
Passive	cannot deliver net energy	R, L, C

Build

1 · Active vs passive

PREDICT

A phone charger's transistor amplifies a signal. Does the transistor create energy, or take it from somewhere?

An **active element can deliver net energy** to the rest of the circuit. A **passive element cannot**:

- R turns energy into heat.
- L and C store energy and give it back, but never more than they took.

An amplifier (transistor, op-amp) is **active**. It does not create energy: it draws it from its DC supply and uses it to make a bigger copy of the input signal.

WORDING IN THE CLASS NOTES

Class p.2 says an active element "doesn't require any external energy source". Read that as: *the element is the thing that supplies energy to the circuit*. It does not mean an amplifier runs without a supply. It doesn't.

CHECK 1

(a) Is a battery being charged acting as active or passive at that moment? (b) Why is an op-amp called active even though it needs ± 15 V to work?

2 · Independent sources

PREDICT

An ideal 10 V source is connected to 1 Ω , then to 1 M Ω . What happens to its voltage and to its current in each case?

- **Ideal voltage source:** keeps its voltage **fixed whatever current flows**. Internal resistance = **0**.
- **Ideal current source:** keeps its current **fixed whatever voltage appears**. Internal resistance = ∞ .

Those two internal resistances return in 05: "switching off" a source means replacing it with its internal resistance, so a V source becomes a **short** and an I source becomes an **open**.



Independent V
fixed V



Independent I
fixed I



VCVS
 $V = \mu V_x$



CCVS
 $V = r I_x$



VCCS
 $I = g V_x$



CCCS
 $I = \beta I_x$

Circle = independent source. Diamond = dependent source. + / - inside = voltage source; arrow inside = current source.

3 · Dependent (controlled) sources

PREDICT

A transistor's collector current is about 100 times its base current. What kind of source would you draw to model that?

A **dependent source** outputs a value set by a voltage or current **somewhere else in the circuit** (class p.2: "value depends on another parameter; it is not constant").

Name	Output	Controlled by	Law	Unit of the constant
VCVS	voltage	a voltage V_x	$V = \mu \cdot V_x$	none
CCVS	voltage	a current I_x	$V = r \cdot I_x$	Ω
VCCS	current	a voltage V_x	$I = g \cdot V_x$	S (siemens)
CCCS	current	a current I_x	$I = \beta \cdot I_x$	none

The class names them by what they **depend on** (set 2 p.2): VDVS = VCVS, VDVS = VCCS, CDVS = CCVS, CDVS = CCCS. Same four sources.

The class homework (set 2 p.2–3): find a real device for each.

Source	Device	Why it fits
VCVS	op-amp (IC 741)	output voltage = gain $\times$ input voltage
CCCS	BJT (BC547)	$I_C = \beta \cdot I_B$
VCCS	MOSFET	drain current is set by the gate voltage: $i_D = g_m \cdot V_{GS}$
CCVS	transimpedance amplifier (the class answer: optical fibre)	turns a photodiode's current into a voltage, e.g. in an optical-fibre receiver

UNITS OF A CCVS

In 06 the source is marked **2I**. A voltage equal to "2 $\times$ a current" only makes sense if 2 carries units of Ω . The pack reads it with $k\Omega$ and mA, so 2I means **2 $k\Omega \times I$** : $I = 1.5$ mA gives 3 V. The class working in set 2 reads it as **2 $\Omega \times I$** instead, and gets a different answer. 06 shows both.

CHECK 2

(a) Draw the symbol for a CCCS with value $3I_1$. (b) Which of the four is the op-amp model, and what is μ called for an op-amp?

4 • The R, L, C laws

PREDICT

If the current in an inductor is constant, what voltage appears across it?

	Resistor R	Inductor L	Capacitor C
Law	$v = iR$	$v = L \cdot \frac{di}{dt}$	$i = C \cdot \frac{dv}{dt}$
Where it comes from	Ohm: $V \propto I$	$\psi = N\phi = Li$, and $v = \frac{d\psi}{dt}$	$q = Cv$; differentiate: $\frac{dq}{dt} = i$
Energy	dissipated as heat	stored in the magnetic field: $\frac{1}{2}Li^2$	stored in the electric field: $\frac{1}{2}Cv^2$
DC steady state	resistor	short circuit ($di/dt = 0 \Rightarrow v = 0$)	open circuit ($dv/dt = 0 \Rightarrow i = 0$)

SLIP IN THE CLASS NOTES

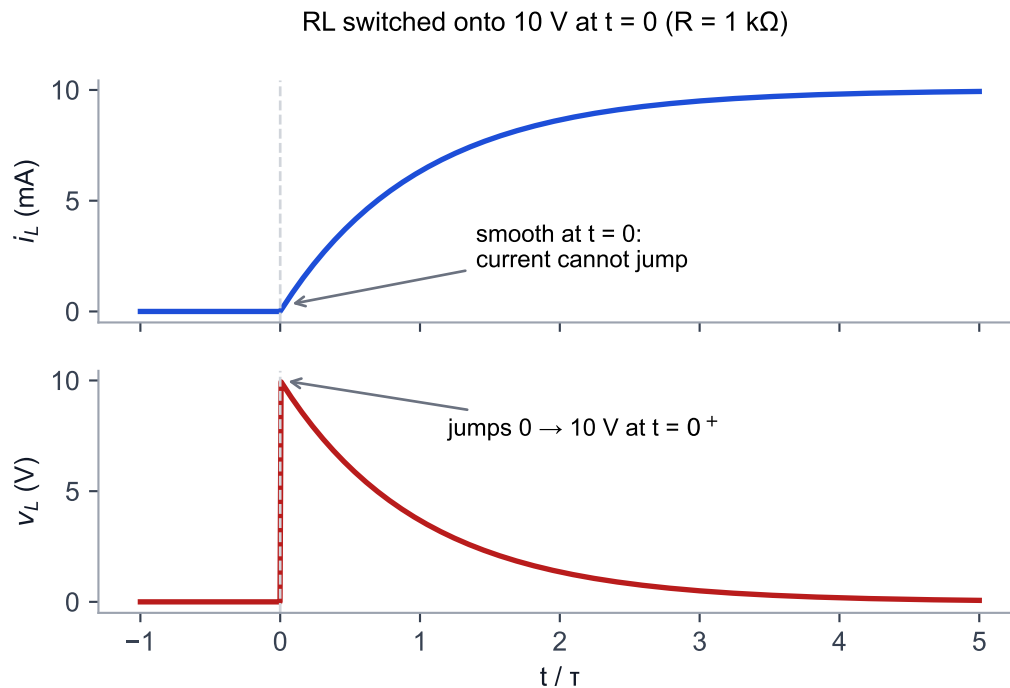
Class p.3 writes the inductor law as " $d/dt (Ci) = C di/dt$ ". It is **L**: $v = d(Li)/dt = \mathbf{L di/dt}$. The side box on the same page ($\psi = Li$) confirms it.

5 • What can't jump

PREDICT

A switch connects 10 V to an R–L circuit at $t = 0$. Just after the switch closes, which one can jump instantly, the inductor's current or its voltage?

- **Inductor current cannot change instantly.** A jump would need $di/dt = \infty$, so $v = L \cdot di/dt = \infty$.
- **Capacitor voltage cannot change instantly.** A jump would need $i = C \cdot dv/dt = \infty$.
- A **resistor** has no memory: its current and voltage can jump (class p.20).



The inductor current rises smoothly from 0. The inductor voltage jumps from 0 to 10 V at the instant of switching.

CHECK 3

(a) Which quantity of a capacitor may jump at $t = 0$: its voltage or its current? (b) Why does a large spark appear when you open a switch in series with a motor coil?

6 · Lumped vs distributed parameters

PREDICT

A 50 Hz power line is 300 km long. A phone's 2.4 GHz Wi-Fi trace is 3 cm long. Which one is "short" compared with the wave travelling along it?

The class split (set 2 p.1): **lumped** parameters are "visible"; **distributed** parameters are "not visible", a property spread along the conductor.

- **Lumped:** each R, L, C sits at one point, as a separate component. Wires between them are ideal. KVL and KCL hold exactly. Every circuit in this pack is lumped.
- **Distributed:** resistance, inductance and capacitance are **spread along the length** of a conductor (per metre), so no single component exists to point at. Transmission lines, antennas and long cables work this way.

When a circuit can be treated as lumped depends on its size compared with the wavelength $\lambda = \frac{c}{f}$:

Signal	$\lambda = c/f$	A circuit of that size is
50 Hz mains	$3 \times 10^8 / 50 = \mathbf{6000 \text{ km}}$	lumped, even across a city
1 GHz	$3 \times 10^8 / 10^9 = \mathbf{30 \text{ cm}}$	distributed at board scale

Rule of thumb: lumped when the circuit is much smaller than λ (often stated as below $\lambda/10$). For the Predict: the 300 km line is $\lambda/20$, so it still counts as lumped (just). The 3 cm trace is about a quarter of $\lambda = 12.5 \text{ cm}$, so it is distributed.

CHECK 4

Why does KVL around a loop become only approximate when the loop is comparable in size to λ ?

Exam form

- **Active element:** can deliver net energy to the circuit (sources; amplifiers, which draw from a supply).
Passive: cannot (R, L, C).
- **Dependent source:** its value depends on another circuit variable. Four types: VCVS, VCCS, CCVS, CCCS (table in step 3, with symbols).
- **Laws:** $v = iR$, $v = L di/dt$, $i = C dv/dt$.
- **Continuity:** $i_L(0^+) = i_L(0^-)$ and $v_C(0^+) = v_C(0^-)$.
- **DC steady state:** $L \rightarrow \text{short}$, $C \rightarrow \text{open}$.
- **Lumped:** parameters concentrated in separate elements, circuit $\ll \lambda$. **Distributed:** R, L, C spread along the conductor (transmission lines), circuit comparable to λ .
- Devices: VCVS = op-amp, CCCS = BJT, VCCS = MOSFET, CCVS = transimpedance amplifier.

Traps

- "An inductor stores current" and "a capacitor stores voltage" (class-note wording) $\rightarrow$ the exam answer is **energy in a magnetic / electric field**.
- Writing C for L in $v = L di/dt$.
- A dependent source's value is **not** a fixed number: it changes whenever its controlling variable changes.

Self-test

1. Name the four dependent sources and give the unit of each constant.
2. Derive $v = L di/dt$ starting from $\psi = N\phi$.
3. Why is an inductor a short circuit at DC steady state?
4. Energy stored in 1 mH carrying 10 mA?
5. True or false: a resistor's current cannot change suddenly.
6. A 10 cm circuit runs at 3 GHz. Lumped or distributed? Show λ .

Answers

Check 1. (a) Passive: it is absorbing energy at that moment. (b) It delivers more signal power to the load than the input provides; the extra comes from its supply.

Check 2. (a) A diamond with an arrow inside, labelled $3I_1$. (b) VCVS; μ is the open-loop (voltage) gain.

Check 3. (a) Its current ($i = C dv/dt$ can jump; v cannot). (b) The coil current cannot stop instantly, so $v = L di/dt$ becomes huge and arcs across the opening contacts.

Check 4. KVL assumes the voltage is the same all along a wire at one instant. When the loop is comparable to λ , the wave takes a real fraction of a period to travel around it, so different points on "the same wire" are at different phases.

Self-test.

1. VCVS (μ , no unit) · CCVS (r , Ω) · VCCS (g , S) · CCCS (β , no unit).
2. $\psi = N\phi \propto i \Rightarrow \psi = Li$; $v = d\psi/dt = d(Li)/dt = L di/dt$.
3. At steady state $di/dt = 0$, so $v = L \cdot 0 = 0$ whatever the current: that is a short.
4. $\frac{1}{2} \times 1 \times 10^{-3} \times (10 \times 10^{-3})^2 = \mathbf{50 \text{ nJ}}$.
5. False: a resistor has no memory.
6. $\lambda = 3 \times 10^8 / 3 \times 10^9 = \mathbf{10 \text{ cm}}$: the circuit is a whole wavelength long, so **distributed**.